

Chandrika Sethumadhavan

Embedded Systems Engineer

ML and sensors enthusiast

[Github](#)

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D.O.B: 06.04.2002, Female



"Turning sensor signals into actionable intelligence through embedded ML"

DO I FULFILL YOUR PROFILE?

- 5+ projects in embedded systems, machine learning, and computer vision. Published in Springer (PCCDA'23).
- Professional experience at Fraunhofer ISE and IMTEK lehrstuhl für Microsystemtechnik in sensor interfacing and ML-based defect detection.
- Applied deep learning techniques (YOLOv8, Mask R-cnn, segmentation, enhancement) for industrial applications.
- Experience deploying ML model using Flask, AWS CI/CD, and cloud-based apps on projects.

Want to get an impression of my work? [click here for projects-masterlist](#)

EDUCATION :

University of Freiburg	M.Sc in Embedded Systems Engineering	GPA: 2.2	2023 – *
Anna University, India	B.E in Electrical, Electronics Engineering	GPA: 1.4	2019–2023

EXPERIENCE :

Master Thesis at IMTEK, Uni Freiburg   Ongoing

- Building a physics-informed ML framework to predict O₂ release from ANT-EPO systems across varying temperature and concentration conditions.
- Integrating kinetic models (Henry's Law, Arrhenius) with signal processing on pressure-sensor data to extract physically meaningful features.
- Comparing Random Forests, XGBoost, LSTMs, and PINNs for prediction accuracy and physical consistency in micro-actuation and drug-delivery contexts.

Work student at ECOSENSE, IMTEK, Uni Freiburg
– November 2025



ECOSENSE

September 2024

- Designed, assembled, and tested PCBs using **Eagle** for environmental sensors; firmware flashing and debugging with **Keil uVision**.
- Developed Python-based MQTT logging for real-time sensor data from The Things Network (TTN) with automatic CSV export and interval-check logic.
- Integrated InfluxDB Cloud and configured Grafana dashboards for live monitoring of sensor values.

Work student at Fraunhofer ISE, Freiburg



December 2023 – June 2024

- Optimized an image processing pipeline for solar panel quality control by integrating feature extraction, edge detection, and thresholding techniques. Implemented defect detection using YOLOv8 and standard algorithms, increasing prediction accuracy up to **99.4%**.
- Utilized **OpenCV**, **PIL**, **NumPy**, and **SciPy** libraries for precise statistical analysis of mesh screen irregularities.
- Experience in handling and collecting data from industrial 3D Measuring Laser Microscope LEXT to optimize the printing of solar cell finger-lines.

TECHNICAL SKILLS :

Programming Languages:

Python, C, SQL

Image Processing & Machine Learning:

Libraries: OpenCV, YOLOv8, TensorFlow (+Lite), NumPy, SciPy, Edge Impulse, scikit-learn, PyTorch, Optuna

Concepts: TinyML, MLOps, AutoML, Model Deployment, Experiment tracking, Hyperparameter Tuning, Data Preprocessing, Model Evaluation, Carbon-Aware ML.

Development Tools:

DevOps/Platforms: AWS (Elasticbean env, EC2 instances), Docker, GitHub, Gitlab, CI/CD, FastAPI

Environments: Flask, VS Code

Embedded Systems:

Worked with: Raspberry Pi, MSP430, ESP32

Skills: Microcontroller Programming, Sensor Integration, PCB Design, Testing & Debugging, Firmware Flashing, Keil uVision

IoT Data & Visualization Tools:

MQTT Protocol (TTN integration), InfluxDB (Cloud Setup, Time-Series DB), Grafana (Dashboard Config, Flux Queries)

PROJECTS :

Meals or Miles: Recipe Carbon Footprint Analyzer - Black Forest Hackathon, Offenburg 2025

[Github](#)

- Built a recipe analysis tool that calculates meal carbon emissions and suggests sustainable ingredient alternatives using vector embeddings for multi-language ingredient matching.
- Implemented multi-objective optimization to identify key emitters (15% of total carbon) and generate nutritionally-balanced replacements, reducing carbon footprint while maintaining similar protein, carbs, and fat profiles.

- Built Flask-based backend with real-time comparison visualization. System operates with zero API calls and minimal computational overhead.

(Coursework) Resource-Aware AutoML for Deep Learning

[Github](#)

- Built a resource-aware AutoML pipeline combining zero-cost proxies and carbon-aware techniques for efficient neural network model selection on vision datasets.
- Integrated CodeCarbon for carbon footprint tracking during hyperparameter optimization using Optuna, minimizing environmental impact.
- Achieved **86.26%** accuracy and **0.796 F1-score** on skin cancer classification dataset using PyTorch and TorchVision.
- Evaluated across multiple open-source datasets (flowers, emotions, fashion, skin cancer) with accuracy ranging from 64.6% to 97.0%.

End-to-End ML Pipeline: Academic Performance Prediction

[Github](#)

- Built a complete ML pipeline from data preprocessing to model training, achieving R^2 scores around **0.88** with multiple ML models like Ridge, Catboost etc.
- Ensured robust logging and exception handling for pipeline stability. Developed a Flask-based web app for real-time predictions. Deployed the model on AWS using CI/CD pipelines for seamless updates. Managed code and version control with Git.

Microcontroller Programming with MSP430

[Github](#)

- Developed embedded systems using the MSP430 microcontroller, including LED toggling, shift registers, ADCs, interrupt handling, and timer configuration.
- Watchdog Timers (WDT) for system reset in case of deadlock, with task management through preprocessor directives for real-time operations.

(Coursework) "Prediflex" Smart Insole - Microsystems Design Lab

[Github](#)

- Virtual design and analytical modeling of an energy-harvesting microcontroller system powered by passive piezoelectric sensors that convert biomechanical energy into usable electrical power.
- The harvested energy supports both the microcontroller and wireless communication modules, all conforming to relevant international standards.
- The system architecture emphasizes ultra-low-power operation, the use of sustainable materials, and seamless integration to enable real-time pediatric gait analysis with predictive assessment of potential abnormalities.

LANGUAGE SKILLS :

German (B1) English (C2)